The plan, step by step: what we measure and why it answers the question

Written 9 November 2026 for Gabriel and Léonard. Assembled from `docs/plan.md`, `docs/stage-c-plan.md`, `docs/protocol.md` and section 6 of `docs/check-in-1-report.md`. Nothing here is new work; it is the argument those documents make, laid out in one place.

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1. The final goal

Does the choice of pretraining corpus change what a molecular property model learns?

The deliverable is a write-up due **1 December 2026**. To make the question answerable rather than merely askable, one thing is held rigidly fixed: CheMeleon's architecture, objective and training budget. Width 2048, depth 6, masked descriptor regression, batch 128, early stopping at patience 50. Only the corpus varies.

That constraint is the whole design. Published comparisons of foundation models vary corpus, architecture, objective and scale simultaneously, so a difference between two of them has four possible causes. Holding three of the four fixed is what turns an observation into an attribution.

Why this is hard, stated up front

Section 6 of the Check-in 1 report read the statistics off the 1 October lecture, and they are sobering:

- On ten MoleculeNet datasets, **six methods including plain Chemprop could not be statistically separated at all**: Friedman $p = 0.194$.
- CheMeleon beat Chemprop by 1.20 mean-rank points on ADMET, **inside** the 1.61 critical difference.
- **The split matters ten times more than the method**: 0.78 AUROC under a random split against 0.54 under a scaffold split, from the protocol alone.

So the effect we are hunting is smaller than the noise in the published work. That governs every choice below: paired comparisons, frozen splits, low-label regimes, and spread reported rather than point estimates. It also means **a null result is a real result**, and the plan says to report it as one.

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2. The logical chain

The five stages are not five topics. They are a chain in which each link makes the next one interpretable.

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A  Do the corpora differ?
   ?   no  -> the project is answered, there was no treatment
   ?   yes -> there is something to measure
   ?
C  How much data does pretraining need?
   ?   gives the size at which pretraining works at all,
   ?   and therefore the size to run D at
   ?
D  Does composition matter at that size?
   ?   the original question, now asked where the answer means something
   ?
E  Where does the gain live?
       mechanism: is it concentrated near the corpus?

B  (independent, cheap) Does distance-to-corpus predict where
   the released checkpoint helps?  Needs no pretraining at all.
```

Three features of that ordering are worth naming.

Stage A could have ended the project. If the corpora were chemically indistinguishable, there is no treatment and nothing downstream could mean anything. Running it first, on CPU, was the cheap way to find out.

Stage D depends on Stage C's answer, not just its existence. D compares three corpora at *one* size. If that size sits below saturation, a difference between corpora is really a difference in effective size, and the comparison measures the wrong thing. Stage C's curve supplies the size. This is also why the low end of the curve matters: if it is already flat at 5,000, every later experiment runs at 5,000 and Stage D becomes cheap.

Stages A, B and E need no GPU. That was deliberate - it means the project survives a bad week in the queue.

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3. Stage by stage

Stage A - Do the corpora differ? (done)

Question	Are PubChem, COCONUT and the two ZINC arms chemically distinct?
Method	Descriptive statistics, scaffold overlap, corpus-to-corpus similarity, task-to-corpus distance, t-SNE. CPU only.
Outputs on disk	runs/analysis/stage_a_corpora/: corpus_stats.csv, corpus_similarity.csv, scaffold_overlap.csv, task_distance.csv, tsne_corpora.png
Answer	Yes, by a wide margin.

Five findings, three of which constrain everything after:

1. The corpora are chemically distinct, not variations on a theme - largest property gap 2.15 standard deviations, histograms overlapping 28 percent.
2. Natural-product-likeness is **entangled** with ordinary properties: selecting on it also selects on saturation, aromaticity, size and stereochemistry. So a later corpus effect cannot be attributed to "naturalness" without that caveat.
3. **Nominal size is not effective size.** One COCONUT molecule in five has a near twin in a 10,000-molecule sample, against one in twenty for PubChem. COCONUT needs about 40 percent fewer spheres to cover the same count. This bears directly on the size curve: 50,000 COCONUT molecules are not 50,000 PubChem molecules.
4. **Corpus identity, benchmark proximity and benchmark exposure are three confounded variables.** PubChem already contains 73 percent of BBBP and 93 percent of ESOL by exact structure. And CheMeleon's corpus is not the uniform random PubChem draw its paper claims - it is enriched roughly hundredfold for well-known compounds.
5. The effect we are hunting is smaller than the published noise.

What Stage A buys is the right to interpret a null: the corpora genuinely differed, so indifference would be a finding about the model, not about the data.

Stage B - Does distance to the corpus predict where pretraining helps?

Question	Does a benchmark's chemical distance from the pretraining corpus predict how much the released checkpoint helps on it?
Method	Fine-tuning only, on CheMeleon's released checkpoint. No pretraining. The x-axis is Stage A's task distances.
Figure	Scatter: task-to-corpus distance against gain over the scratch baseline, one point per benchmark

Cost	Fine-tuning only - the cheapest stage with a GPU result
Status	Blocked on the fine-tuning loop, and on nothing else

This is the stage that should already be done. It needs no pretraining, no corpus preparation and no size curve - only the downloaded checkpoint and a working fine-tuning loop.

A caution from the call, recorded in the plan: a molecule's distance to the *pretraining corpus* is not its distance to the *labelled training set*, and the two do not track each other.

Stage C - How much pretraining data is needed? (running)

Question	What is the smallest PubChem corpus at which descriptor pretraining still beats Chemprop trained from scratch?
Method	Re-pretrain on nested PubChem subsets at 500, 1k, 5k, 10k, 25k, 50k, 100k, two seeds each. 14 conditions. Fine-tune each on BBBP and ESOL over five frozen scaffold splits and four label counts.
Figure	The headline figure. Downstream score against corpus size, with error bars from two seeds, baselines as horizontal lines, and the released 944k checkpoint as the top point.
Also produces	the operating size for Stage D
Status	13 of 14 pretrained, 63 GPU-hours. Fine-tuning not started.

Four design choices that make the curve mean something:

- **Nested subsets.** The 500-molecule corpus is a prefix of the 1,000, which

is a prefix of the 100,000. So bigger corpora *add* molecules rather than resampling, and the curve is a curve in one variable.

- **Early stopping, not fixed epochs.** Fixing epochs would give the 10k corpus

ten times fewer gradient steps than the 100k one, mixing corpus size with training budget. Fixing steps would make the small corpus overfit. Training to convergence is the only rule fair to both. The epochs reached are reported.

- **One frozen objective.** All fourteen conditions standardise their descriptor

targets against the 100k condition - 1,562 usable targets of Mordred's 1,613. Standardised against themselves, the smallest would fit 1,416 targets and the largest 1,562, and the curve would measure the objective widening at the same time as the corpus growing.

- **Two baselines, not one.** A size-matched randomly initialised model

attributes any gain to pretraining rather than capacity - this is the control CheMeleon's own paper ran. A well-tuned Chemprop at its own configuration is the harder line and the practitioner-relevant one. The first is easy to beat; the second is what the lecture found statistically tied with the best foundation models.

Stage D - Does composition matter at a fixed size?

Question	At one corpus size, does it matter whether the molecules come from PubChem, COCONUT or NP-like ZINC?
Method	Three corpora at the size Stage C shows is on the flat part. Three seeds.
Figure	Paired differences between the three arms at matched size, with intervals
Status	Not started; waits on Stage C's curve

This is the original proposal, now asked in a place where the answer is readable. Two cautions carry over from Stage A:

- **Matched nominal size is not matched effective size** (finding 3). 50,000

COCONUT molecules carry more redundancy than 50,000 PubChem ones.

- **Benchmark exposure differs by construction.** At 50,000 molecules the three

arms carry roughly 127, 85 and zero benchmark structures. A PubChem win would be unreadable: better chemistry, or 127 test structures already seen? The protocol's recommendation is to run Stage D with `--exclude-benchmarks` on all three arms, at the cost of no longer being directly comparable to the Stage C curve, which keeps its overlap. Both numbers get recorded.

Stage E - Where does any gain live?

Question	Does the gain concentrate in test molecules chemically close to the pretraining corpus?
Method	Bin test molecules by similarity to the pretraining corpus; compute the gain per bin. No new training.
Figure	Gain against similarity-to-corpus bin
Status	Not started; runs on whatever has been fine-tuned

This is the mechanism stage. A corpus effect that concentrates in near bins is evidence of transfer of relevant chemistry; one spread evenly across bins is evidence of something more generic.

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4. The figures, collected

#	figure	stage	exists?
1	t-SNE of the four corpora, plus property gap and scaffold overlap tables	A	yes
2	Task-to-corpus distance against released-checkpoint gain	B	no
3	Downstream score against pretraining corpus size, with baselines and the released checkpoint	C	no
4	Low-label curves: score against labelled-set size, per condition	C	no
5	Three corpora at matched size, paired differences	D	no
6	Gain by similarity-to-corpus bin	E	no

Figure 3 is the headline. Figure 1 is the only one that exists. Figures 2 through 6 all require the fine-tuning loop.

Figure 4 deserves more weight than its position suggests. The published evidence says the pretraining advantage is largest below a few hundred labelled molecules and roughly halves by several hundred - about 0.07 AUROC at 60 BACE molecules falling to 0.04 at 847. BBBP at 1,631 training molecules and ESOL at 902 are both near the top of the range where the effect is smallest. **The 50- and 100-label columns are where a signal can plausibly exist**, which is why the protocol fixes label counts of 50, 100, 500 and all.

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5. How it adds up

Read the stages as one argument:

1. The corpora differ, measurably and in ways that are not reducible to one axis (**A, established**).
2. Pretraining on PubChem beats training from scratch, and here is the corpus size at which that begins

and the size at which it stops improving (**C, pending the y-axis**).

3. At a size on the flat part, swapping PubChem for COCONUT or NP-like ZINC changes the downstream score by this much (**D**).

4. The gain, where it exists, is concentrated in test molecules close to the corpus - or it is not (**E**).

5. Independently, a benchmark's distance from the corpus predicts how much the released checkpoint helps on it - or it does not (**B**).

Together those say whether the corpus matters, how much data is needed before the question is even well-posed, which corpus wins at matched size, and by what mechanism. That is a complete answer to the project's question, including if the answer is "it does not matter much" - which Stage A's rigour is what makes publishable rather than merely inconclusive.

What would make it not add up

Worth stating so it can be guarded against:

- **If the scratch baseline is misconfigured**, the whole curve floats against

an arbitrary line. `docs/protocol.md` does not currently pin the baseline's architecture. It should name both baselines explicitly.

- **If splits differ between conditions**, nothing is comparable. Already

guarded: five scaffold splits per benchmark are frozen on disk and every run reads them.

- **If Stage D runs below saturation**, it measures size while claiming to

measure composition.

- **If benchmark exposure is not handled**, a PubChem win in Stage D is

uninterpretable.

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6. Where we actually are, 9 November

stage	state
A	done, five findings, figure and tables on disk
B	blocked on the fine-tuning loop, and on nothing else
C	13 of 14 checkpoints pretrained, 63 GPU-hours; evaluation not started
D	not started, waits on C
E	not started

The last pretraining run, `pubchem_n100000_s0`, is at epoch 249 on its third attempt after two preemptions; its seed-1 twin stopped at 264.

Four of the five stages run through `scripts/finetune.py`, which is 36 lines ending in `raise NotImplementedError`. It is blocker B2, assigned to Léonard, two days, due to begin 10 November - so it is not late. But pretraining has run roughly nine days ahead of schedule, which means that file is now the only thing on the critical path, and Stage B has been waiting on it without needing anything else.

Two things that can be done before it lands, neither needing a GPU:

- **Pin both baselines in `docs/protocol.md`** under Amendments, which

currently reads "None yet". Do it before task 7 runs, not after.

- **Add the Chemprop-plus-descriptors baseline** to the task list. It is on

nobody's. Adding computed descriptors to Chemprop is worth about as much as pretraining is - roughly 0.09 AUROC on BACE at 60 molecules - and since CheMeleon's pretraining target *is* computed descriptors, without it a corpus effect and a descriptor effect cannot be told apart.